

Treatment and Prevention of Scabies

There have been numerous reports of the effectiveness of **ivermectin** in the treatment of scabies. The usual dosage is $200\ \mu\mathrm{g}/\mathrm{kg}$; often, the dose is repeated in 10 to 14 days, although the optimal dosage schedule for scabies has not been definitively established. A clinical study comparing ivermectin to topical 5% permethrin showed that a single dose of ivermectin provided a cure rate of 70%, which increased to 95% with a second dose given 2 weeks later. In contrast, a single application of permethrin was curative in nearly 98% of patients. A topical preparation of 1% ivermectin has shown efficacy in early clinical trials but is not yet commercially available. In cases of **crusted scabies**, combination therapy with oral ivermectin and a topical anti-scabicide is commonly used due to the high mite burden, and multiple treatment courses may be necessary to eradicate the infestation.

Side effects of ivermectin are generally mild and transient. Rare adverse events include hypotension, laryngeal edema, and encephalopathy. One case report suggested an increased death rate among elderly nursing home patients treated with ivermectin; however, subsequent studies have not confirmed this association.

Prevention

To prevent transmission, close contacts of an infected individual should be treated with a topical scabicide. Treatment is important even in asymptomatic individuals, as they may be harboring mites during the incubation period. To prevent reinfestation via fomites, bed sheets, pillowcases, towels, and clothing worn in the past 5 days should be washed and dried using the hot cycle or dry cleaned. Since the scabies mite can survive up to 3 days off the human body, carpets and upholstered furniture should be vacuumed thoroughly. Pets do not require treatment, as they do not carry the human scabies mite (*Sarcoptes scabiei* var. *hominis*).

Box 208-2: Treatment for Scabies

Drug	Dose	Comments
Permethrin 5% cream	Apply for 8–14 hours; repeat in 7 days	First-line treatment in the U.S.; pregnancy category B
Lindane 1% lotion	Apply for 8 hours, then wash off; repeat after 1 week	Not recommended for children <2 years, pregnancy, or lactation; resistance increasing; banned in California
Crotamiton 10% cream	Apply on 2 consecutive days; repeat within 5 days	Has antipruritic effects; may be less effective than other agents
Precipitated sulfur 5%–10%	Apply for 3 days, then wash off	Safe for infants <2 months and during pregnancy; messy application; limited efficacy data
Benzyl benzoate 10% lotion	Apply for 24 hours, then wash off	Not available in the United States
Ivermectin, 200 $\mu\text{g/kg}$	Single oral dose; may repeat in 10–14 days	Highly effective and safe; can be combined with topical agents, especially in crusted or resistant cases

Other Mites

While the scabies mite is the most well-known mite affecting humans, other mites can also cause disease. Mites belong to the class **Arachnida**, order **Acarina** (which includes ticks), and, like scabies mites, are eight-legged organisms with a fused cephalothorax.

Clinically significant mites are generally classified into three suborders: **Trombidiformes**, **Mesostigmata**, and **Sarcoptiformes**. They are often categorized based on their natural host or habitat.

Follicle Mites

Demodex species are common commensals of the human pilosebaceous unit. These mites inhabit hair follicles on the face, scalp, and upper chest. *D. folliculorum hominis* resides within the hair follicle, while *D. brevis* prefers the sebaceous gland infundibulum. The prevalence of **Demodex** infestation increases with age.